

CS 455

Program #5

Ray Tracing

Dr. Egbert

For this program you will write the beginnings of a ray-tracer. This ray tracer will have basic functionality, but we will add more functionality in the next Program. In addition, you are certainly welcome to add more advanced features, if you wish. For this program, your ray tracer must have the following capabilities:

1. The ability to ray trace spheres.
2. The ability to trace one ray per pixel.
3. At least one directional light source in the scene.
4. A decent illumination model for intensity calculations, allowing for ambient, diffuse, and specular components.

We will give you two test scenes that you need to render. In addition, you will need to render a scene that you create which is more complicated than the ones we give you.

You are welcome to write this in any language you want. You won't need to use OpenGL for this program, and in fact you won't need any graphics language. You should trace the rays, then output the color values for the pixels into a file.

Probably the simplest approach is to output to a .ppm file. A .ppm file has the following format:

```
Magic Number
# of columns  # of rows
Maximum color value
R G B  R G B  R G B  . . .
```

where the magic number is: P3 for ASCII RGB, P6 for binary